Title: Benchmarking Simple Machine Learning Models for COVID-19 Severity Prediction on Single-cell Data

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Abstract

In this study, we benchmarked a suite of simple supervised learning models (Logistic Regression, Linear SVM, Random Forest, XGBoost, and KNN) to predict COVID-19 severity based on single-cell training data. By focusing on immune cells and training on multiple folds, these models achieved high performance across standard metrics, with F1-scores averaging 0.81 and AUROC scores reaching 0.930.05. We also trained Topic-Based Random Forest models, using only 10 latent features, which remained competitive (F1 = 0.77). We identified important gene biomarkers that signal COVID-19, and demonstrated that simple machine learning models are capable of predicting disease severity on single-cell data.

Introduction

Predicting disease from cell data

Single-cell RNA-seq and other cell omics data can be used to predict disease presence, subtype, stage, drug response, etc. [1]. Single-cell data can be complicated to predict on, given the nature of high-dimensionality and non-linearity present in the data features. Basic linear models (linear regression, decision trees, SVM) and ensemble models (random forest, XGBoost, simple neural networks) are among the most accessible and easy to apply to data, in general. [2]

However, for prediction right now, there are many more complicated models and foundation models trained on very large sets of data. Graph attention networks, gradient boosting, and attention-based neural networks can accurately distinguish disease from healthy states and predict progression or immunity from single-cell data [3, 4, 5]. Frameworks like scFeatures summarize thousands of cells into sample-level features for prediction of disease [1] . Editorial concepts such as scDisPreAI propose pipelines that integrate multi-disease, multi-stage single-cell datasets with AI for clinical decision support and biomarker discovery [6]. This is not a comprehensive list; there are many more complex models that predict outcomes from cell data.

Complex machine learning models such as deep learning systems and foundation models achieve strong performance but have important limitations. They take extreme amounts of data, compute, and storage to train. Their decisions are also hard to reliably calibrate or validate under distribution shift, and deep networks are known to exploit shortcuts that fail in new settings [7, 8].

On the other hand, simple, supervised machine learning models such as Logistic Regression, Linear SVM, Random Forest, Topic-based Random Forest, and XGBoost are portable and easy to configure on any amount of data. These models take relatively little compute and storage, and are already used in many clinical risk models [9, 10].

Aims

In this project, we aim to train these simple machine learning models on single cell data and 1) determine whether or not they are capable of predicting disease in a high dimensional space and 2) compare the performance between different simple models trained on the same data.

In the following, we trained lightweight models on cell-data, and then reported performance metrics.

Methods

We predicted on the Stephenson et al. (2021) COVID-19 peripheral blood mononuclear cell (PBMC) atlas, a single-cell PBMC dataset with 780k high-quality cells from 140 samples, E-MTAB-10026, accessible here at CellxGene [11]. This dataset covers healthy and COVID-19 patients.

Figure 1: Participant groups and sample sizes in E-MTAB-10026 [11]

Group (blood PBMC)	**Number**	**Notes**
Healthy controls	24	No COVID-19
Asymptomatic	12	PCR+ no symptoms
Mild	26	Ward, no oxygen
Moderate	32	Ward, oxygen
Severe	15	ICU and/or non-invasive ventilation
Critical	17	Intubated/ventilated
Non-COVID severe respiratory illness	5	Hospitalized controls
IV-LPS (acute inflammation control)	12	Healthy volunteers, 90 min & 10 h post-LPS

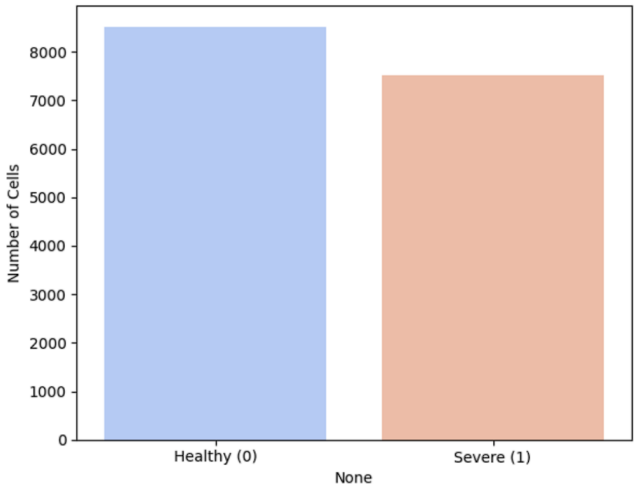
Subsampling the Dataset

We subsampled this dataset to have clearer predictions, with several limitations. We only kept cells for patients with Healthy, Severe, or Critical status labels to find more distinct gene programs. We chose to discard mild and moderate cases to reduce noise and predict for binary outcomes. To focus analysis on COVID-19 immune responses, non target cells (red blood cells, platelets, erythrocytes, malignant B cells) [12] were filtered out. When selecting random cells to make up our sample dataset, we capped the contribution from each donor at 500 cells, limiting the contribution of any individual donor. We then applied gene filtering ($\texttt{min_cells} = 50$) and library-size normalization ($\texttt{sum} = 10,000$) followed by log-transformation ($log(1 + x)$) to squash variance; some genes are expressed at much higher counts than others across the cell data. [13, 14] Lastly, we picked only the top 2,000 highly variable genes (HVGs) using Seurat dispersion-based method. This gives us a higher chance to find features correlated to outcomes.[13, 15]

After sub-sampling, we checked and saw that cells were evenly proportioned between being from healthy and severe/critical patients:

Figure 2: Distribution of Clinical Labels

Label	Count	Prop (%)
Healthy	8524	53.15
Severe	7515	46.85



We also checked the number of cells donated by individual donors and their clinical status. There was a very similar distribution between healthy and severe donors:

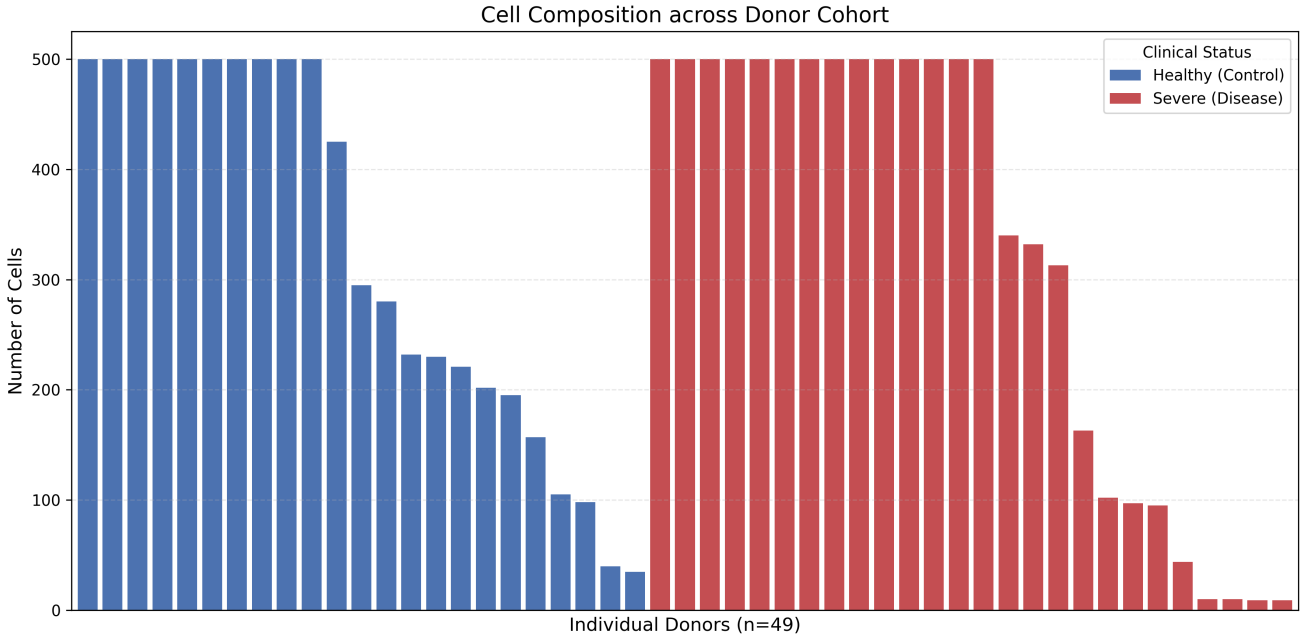


Figure 3: Similar distributions between healthy and severe donor cohorts

Training and Testing

For best results, we would have tested in batches for train data, while leaving out one patient for testing predictions (Leave-One-Group-Out) [16]. However, because we had a weak machine and limited

memory, we used 5-fold GroupKFold cross-validation. This still ensured that no donor was shared between training and test sets, mitigating donor-specific bias in our predictions. The overall runtime to train all models once sat at ~ 45 minutes. We ran the test and train program 3 separate times, each trial yielded very similar model performance (less than 0.01 difference in F1-score).

On each fold, we trained and tested on a suite of linear (Logistic Regression, SVM)and ensemble-based (Random Forest, XGBoost) models, plus KNN. The pipeline was implemented using Scikit-learn and XGBoost frameworks. The logistic regression and linear SVM models were optemized via a SAGA solver with balanced class weights. Using Latent Dirichlet Allocation (LDA) for feature extraction, we extracted 10 discrete gene programs (topics) from a 2,000 gene feature space, allowing us to benchmark a Topic-Based Random Forest. For KNN, we set k=5 (we also tried k=10, 20, 100).

We bench-marked model performance using F1-score and Area Under the Receiver Operating Characteristic (AUROC) curve. We also highlighted the top predictive features for each model. The full code is available here.

Results

The following is a table of our observed F1-scores of each model. Precision and recall are based on predicting 'Severe':

Figure 4: Model performance scores

Model	**F1-Score**	**Precision**	**Recall**
Linear SVM	0.8266	0.8779	0.7874
XGBoost	0.8219	0.9006	0.7702
Logistic Regression	0.8207	0.9100	0.7636
Random Forest	0.8130	0.8741	0.7701
Topic-Based RF	0.7728	0.8258	0.7409
KNN	0.1805	0.8250	0.1063
Average Minus KNN	0.8110	0.8776	0.76644

The average F1-score (minus KNN) is 0.8110. The best performing model measured by F1-score is Linear SVM at 0.8266 and the lowest is Topic Based Random Forest at 0.7728. The overall range between the largest and the smallest is close, and all models perform decently well. XGBoost and Logistic regression have moderately better precision relative to their peers, with about 0.03 advantage. Recall for all models is around the same range, with Linear SVM having the best recall at 0.7874 and Topic Based Random Forest having the lowest recall at 0.8258.

Note that KNN is excluded from this analysis, as it has competitive precision at 0.8250, but very low recall at 0.1063.

ROC Curve

Below, we have plotted the mean true positive rate against the mean false positive rate for each model, known as the Receiver Operating Characteristic (ROC). This statistic is an average of the performance over 5 folds.

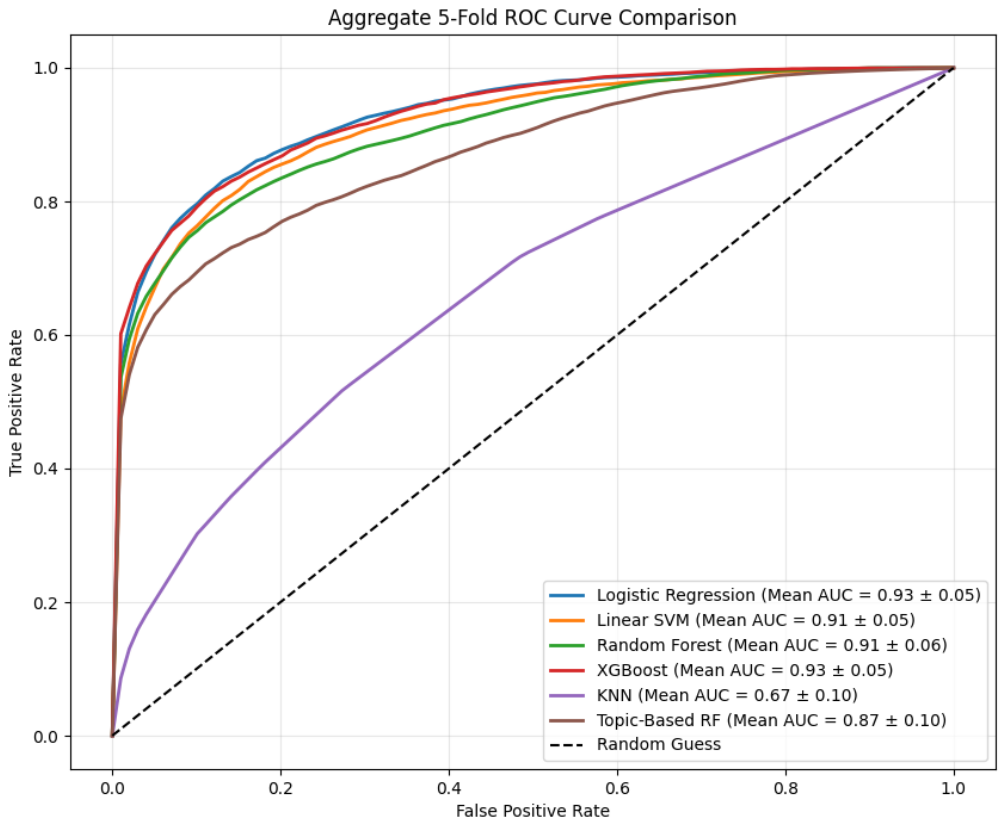


Figure 5: AUROC Curve

We can see that all models except for KNN have excellent AUC scores (Area Under the Curve) at around 0.9. XG Boost and Logistic Regression had slightly higher AUC scores at 0.93, while Topic Based RF performed slightly worse at 0.87. KNN is an outlier with an AUC of 0.67, but it still performs slightly better than random guessing.

Top Predictive Features

Among the four models that predicted off single genes (Logistic regression, Linear SVM, Random Forest, XGBoost), the Clusterin gene was among the highest weighted gene for linear models and the second highest weighted for ensemble models. The HLA_DQB1 gene was an important negatively weighted feature for Logistic regression and Linear SVM, but was noticeably unimportant for Random

Forest and XGBoost.[1]

[1]Note that certain features are represented by their Ensembl IDs (start with ENSG), as they were not mapped to labels in the code. ENSG00000276216 correlates to an RNA gene in the lncRNA class, https://www.genecards.org/cgi-bin/carddisp.pl?gene=ENSG00000276216.

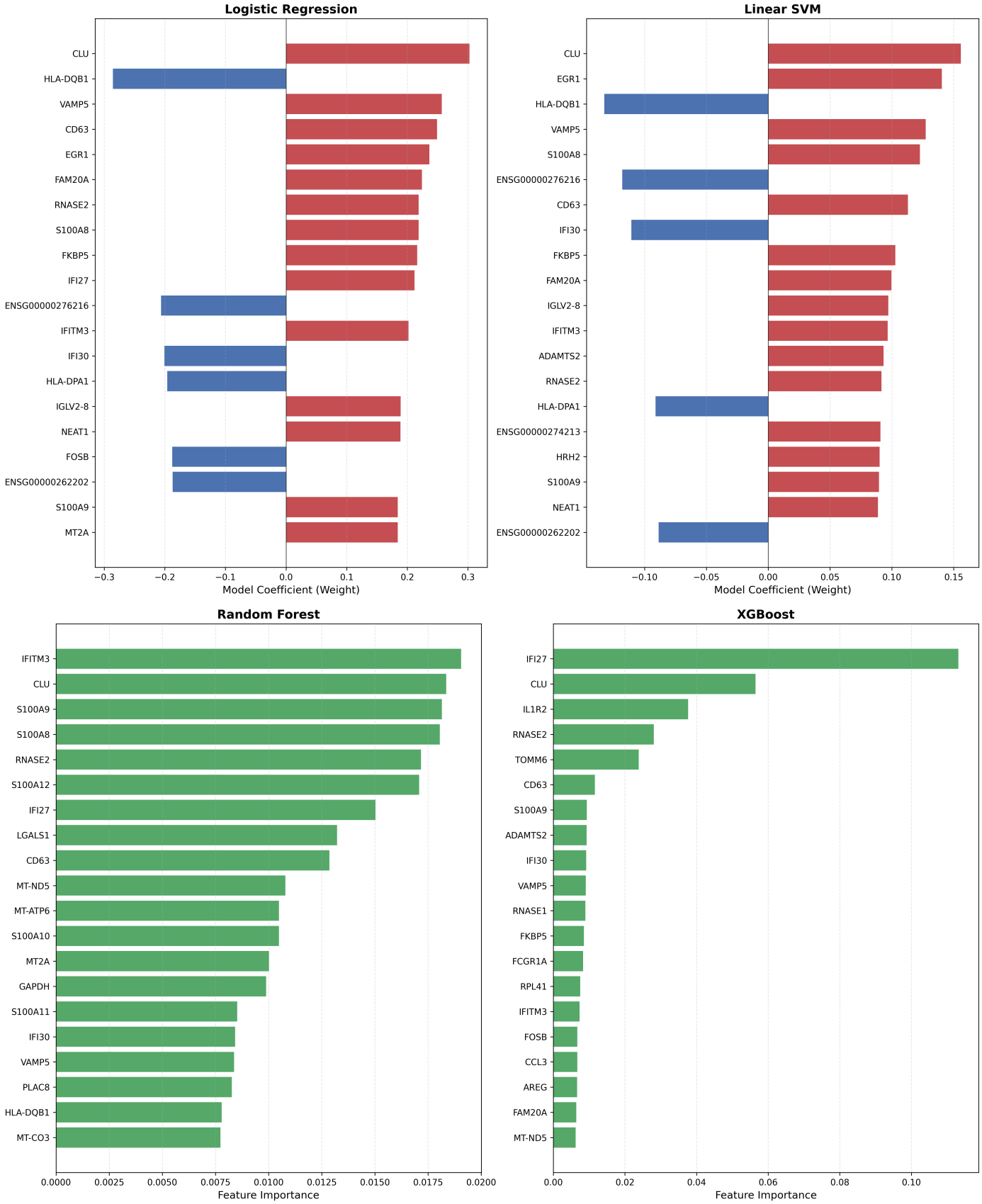


Figure 6: Most impactful predictive features

Discussion

Overall, these simple models were capable of predicting COVID-19 in patients based on cell data. Across the standard classifiers, with F1-scores in a narrow band (0.77–0.83) and an average F1-score of 0.81. These scores are within the range of other real-world ML applications, but below some clinical prediction studies that use deep neural networks.

Figure 7: Representative F1-scores from complex models

Task / Model type	**Best F1-score**	**Training Data**	**Citations**
Cancer outcomes, single-cell foundation model	0.97	95,000,000 cells	[17]
NCS lung cancer, Deep graph convolutional network	0.937	1900 RNA-seq samples	[18]
MV ICU mortality, CatBoost	0.747	2500 patients, 32 features	[19]
CVD, deep CNN with enhanced SMOTE	0.87	N/A	[20]
Coronary disease, hybrid deep + KNN	0.8782	7,291 patients, 17 features	[21]

The high F1-score obtained by these models shows that simple models can capture COVID-19 signal in cell data. Additionally, it suggests that cell data is linearly separable to predict COVID-19, even in extremely high dimensionality. More complex models may be able to find complex signals; on chest-Xray images, deeplearning and fused deep models (e.g. CCSHNet, UNet+Xception) achieved high F1-scores (97%-99%) on COVID-19 classification tasks [22, 23].

Although the topic model was the worst performing model (F1-score: 0.7728), it performed competitively given that it reduced 2,000 genes to just 10 latent topics. [24] With more analysis, we could expect to find that the predictive signature for COVID-19 is not scattered randomly, but rather within some specific gene programs. [25]

The KNN model predicted with good precision (0.8250) but had very low recall (0.1063). This is to be somewhat expected in a high dimensional space, as the euclidean distance over 2,000 features is likely uninformative. Additionally, as we used unseen donors for our cross-validation split, local cell to cell similarity may also have been uninformative (cells could cluster by individual donor rather than disease level).

ROC Curve

AUC score is one metric that summarizes the ability for our models to distinguish between positive vs. negative cases (1.0 = perfect, 0.5 = random).

Figure 8: AUC scores

Model	**Mean AUC $\pm$ SD**
Logistic Regression	0.93 ± 0.05
XGBoost	0.93 ± 0.05
Linear SVM	0.91 ± 0.05
Random Forest	0.91 ± 0.06
Topic-Based RF	0.87 ± 0.10
KNN	0.67 ± 0.10

An AUROC of 0.9 - 0.93 is comparable to high performance clinical ML models for COVID-19 prediction [26, 27, 28]. Given the similarity of these AUROC scores, there may not be any advantage to using more complex models like XGBoost relative to Logistic Regression [27]. The score for Topic Based RF is 4 points lower than other models, again reflecting that Topic Based RF is loses some level of COVID-19 signaling.

Top Predictive Features

The top predictive features were all genes (which is a good sign, we did not want to predict on other labels). Given that the models perform so similarly, it is interesting that they did not select on the same features (other than Clusterin, which is present in the top 2 of all models). This could suggest that genes are strongly linked with each other, or that there are many features that can independently signal COVID-19 with strength. The linear models rely heavily on the presence of HLA-DQB1 in the negative case, while this gene is not in the top 20 of the ensemble models. This could be because the linear models are greedy for consistent downward signals, while this signal might be diluted across many trees.

Figure 9: Top 10 Predictive Features of each model

(a) Log. Reg.

Feature	**W**
CLU	.30
HLA-DQB1	-.28
VAMP5	.25
CD63	.24
EGR1	.23
FAM20A	.22
RNASE2	.21
S100A8	.21
FKBP5	.21
IFI27	.21

(b) Lin. SVM

Feature	**W**
CLU	.15
EGR1	.14
HLA-DQB1	-.13
VAMP5	.12
S100A8	.12
ENSG...	-.11
CD63	.11
IFI30	-.10
FKBP5	.10
FAM20A	.09

(c) Rand. Forest

Feature	**S**
IFITM3	.019
CLU	.018
S100A9	.018
S100A8	.018
RNASE2	.017
S100A12	.017
IFI27	.015
LGALS1	.013
CD63	.012
MT-ND5	.010

(d) XGBoost

Feature	**S**
IFI27	.115
CLU	.058
IL1R2	.038
RNASE2	.028
TOMM6	.023
CD63	.011
S100A9	.010
ADAMTS2	.009
IFI30	.009
VAMP5	.009

We can also examine the biological interpretation to see if the top predictive features of our models make sense with the current literature. Overall, the listed genes are strongly supported as COVID-19 immune and severity markers. CLU, RNASE2, CD63, and IL1R2 are often altered in COVID-19 transcriptomes, and play a role in immune response [29, 30]. IFI27, predicted on by XGBoost, specifically

appears in multiple models and is a inferon-stimulated gene, used as a predictor of COVID-19 severity in blood and lung tissue. [31, 32]. We can also see S100A8, S100A9, S100A12 in all models, which are linked to severe COVID-19 and poor prognosis. [33, 34] IFITM3 are also type 1 inferon response genes, used as diagnostic markers for COVID-19 [33, 35]. Multiple studies show that the negative weighted HLA-DQB1 gene is associated with COVID-19 severity and vaccine antibody responses [36, 37]. Lastly, VAMP5 is an intrinsic defense factor for embryonic stem cells against coronaviruses [38].

While biological interpretation is not the main goal of the study, we can see that these models are predicting on signals that are cannonically aligned with literature, and are effective in general.

Limitations and Future Work

While we have shown that simple models demonstrate strong predictive power, several limitations remain. First, we subsampled our data to be highly binary and restricted to certain cell types. This is not representative of the general population, and we do not know how effective our models are in predicting mild cases. We also reduced the dimensionality; 2,000 features is a large space, but not as large as the original dataset. Also, these models by themselves do not offer any clinical interpretability, so we cannot be confident about the role that highlighted genes play in predicting outcomes.

Future work should train complex and large foundational models on the COVID-19 atlas, for direct comparison with the results in this paper. Also, it could extend the same methods on different cell data to verify performance. Also, integration with clinical metadata (age, history, etc.) could perform very strongly, given that some linked models predict only on metadata.

Conclusion

In this paper, we have demonstrated that simple machine learning models are effective at predicting disease severity from single-cell data. While complex models are being developed at record pace, researchers could consider the strength of using lightweight models and applying them en-masse, given their portability, to see what relationships may be discovered. Sometimes, relationships are linearly identifiable.

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